CHURNGUARD / BUSINESS ANALYTICS

Customer Churn Prediction and Retention Strategy

Milestone 2 | Model improvement, evaluation and business application

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A decision-focused assessment of how the project progressed from the initial XGBoost proof of concept to a validated Extra Trees model and a practical customer-retention workflow.

Executive summary

Customer churn creates a recurring commercial risk: when customers leave, the business loses future revenue and must spend to replace them. Milestone 1 established the scale of the problem and demonstrated that customer information could be used to identify accounts requiring attention. Milestone 2 advances that work by testing a wider range of predictive methods, improving their settings and evaluating a selected model on data not used to choose it.

The selected approach is a tuned Extra Trees classifier. It achieved an F1 score of 93.69% during five-fold cross-validation and 94.86% on a separate test group of 2,252 accounts. On that test group, it identified 351 of 379 customers who actually churned, missed 28 and incorrectly flagged 10 customers who remained. These results indicate strong predictive targeting in the evaluated sample; they do not yet demonstrate that a retention campaign will prevent churn.

Management implications

- Use risk scores to prioritise service attention. The model can help staff concentrate on customers with greater estimated risk rather than treating all accounts alike.
- Preserve a low-cost first response. Incorrect alerts are relatively uncommon in the test sample, but any incentive or outreach still has a cost.
- Recalculate commercial exposure. The revenue-at-risk figures from Milestone 1 were generated using XGBoost and should not be represented as outputs of the new Extra Trees model.
- Approve a measured pilot. Compare targeted outreach with normal service, track costs and determine whether the intervention retains additional customers.

Recommended decision

Proceed with a controlled retention pilot using the Milestone 2 model as a decision-support tool, subject to data-governance checks, clear accountability for follow-up and agreed financial success criteria. A wider rollout should depend on measured customer and commercial outcomes, not model accuracy alone.

1. Business context and continuity with Milestone 1

The first ChurnGuard report, Customer Churn Prediction & Retention Strategy, framed churn as a business issue and introduced an XGBoost-based early-warning application. It reported an observed churn share of approximately 16.8%. This imbalance is important: a system can appear accurate simply by predicting that most customers will stay, yet fail to identify the people who need retention support.

Milestone 1 also demonstrated a concentration of estimated revenue exposure. In its evaluated group of 2,252 accounts, 372 High/Critical-risk accounts represented 16.52% of accounts but 78.60% of probability-weighted monthly revenue exposure. This finding established a useful business principle: targeted attention may be more efficient than a blanket retention campaign.

Those historical results are retained here as the foundation of the business case. They are not presented as recalculated Milestone 2 results. The earlier report also identified account segment, payment method,

tenure, marital status, complaints and login device as leading XGBoost prediction signals. Their importance describes patterns used by that model, not proven causes of customer departure.

What changed between milestones

Dimension	Milestone 1	Milestone 2
Main question	Can customer churn be identified and prioritised?	Which approach provides stronger validated targeting, and how can it be used repeatedly?
Model	XGBoost	Tuned Extra Trees selected from nine candidates
Business output	Risk bands, exposure estimates, prediction app	Model comparison, independent test, repeatable scoring and alert workflow
Evidence gap	Campaign outcomes not yet measured	Campaign outcomes still require a controlled pilot

2. Analytical approach and modelling discipline

Milestone 2 used 11,260 customer records. The analysis reserved 2,252 records (20%) as a final test group and used the other 9,008 records (80%) for development and model selection. The split preserved the proportion of customers who churned in both groups. This matters because the test sample should resemble the overall churn problem while remaining separate from model selection.

Customer identifiers were excluded as predictors. An account number distinguishes customers but does not, by itself, explain their behaviour. Data preparation was incorporated into the modelling pipeline so that the same transformation rules can be applied during training and future predictions. This helps avoid discrepancies between analysis and day-to-day use.

How the comparison was made

Nine modelling methods were evaluated using stratified five-fold cross-validation. The development data was divided into five portions; each model was repeatedly trained on four portions and checked on the remaining one. The resulting scores provide a more stable basis for selection than a single development split. A randomised search tested alternative model settings, with up to 20 combinations per algorithm.

The primary selection measure was the F1 score, which balances recall (finding customers who leave) with precision (avoiding unnecessary alerts). Accuracy remained useful, but it was not the sole decision criterion because churn affects a minority of customers.

3. Model comparison and improvement

The comparison challenged the assumption that the initial XGBoost approach should remain the preferred solution. Different algorithms respond differently to customer patterns; evaluating several families of methods makes the selection more defensible.

Model	Baseline F1	Tuned F1	Change
Extra Trees	0.9316	0.9369	+0.0053
K-Nearest Neighbours	0.8118	0.9232	+0.1114
XGBoost	0.9086	0.9110	+0.0024
Support Vector Machine	0.6535	0.9026	+0.2491
Random Forest	0.8847	0.8849	+0.0002
Decision Tree	0.8194	0.8194	0.0000
Gradient Boosting	0.6862	0.7223	+0.0361
Logistic Regression	0.5365	0.5370	+0.0005
Naive Bayes	0.5007	0.5007	0.0000

All figures are development-stage F1 scores reported for Milestone 2. Changes are absolute score differences, not relative percentage improvements.

Interpretation

Extra Trees ranked first both before and after tuning. Its F1 score increased from 0.9316 to 0.9369, a gain of 0.0053 points. This is a modest improvement on an already strong starting point, not evidence of a dramatic transformation. By contrast, the Support Vector Machine and K-Nearest Neighbours benefited substantially from tuning but still finished below Extra Trees.

XGBoost remained competitive but did not lead this particular validation exercise. Extra Trees was selected because it offered the highest measured balance of finding churners and avoiding incorrect alerts across the nine candidates. The available results do not establish that its advantage would persist in every future customer population, nor that the small tuning gain is statistically significant.

4. Independent test performance

After model selection, the tuned Extra Trees pipeline was assessed against the reserved 2,252 customer records. This is the strongest technical evidence in the milestone because those records were not used to select the winning model.

Measure	Result	Business interpretation
Accuracy	98.31%	About 98 out of 100 outcomes were classified correctly.
Recall	92.61%	About 93 out of 100 customers who left were identified.
Precision	97.23%	About 97 out of 100 customers flagged as likely to leave did leave.
F1 score	94.86%	Strong combined performance on the two targeting priorities.
PR-AUC	98.92%	Strong ability to rank churners where churn is the minority outcome.
ROC-AUC	99.73%	Strong separation between customers who leave and those who stay.

What happened to the 2,252 customers in the test?

Actual customer outcome	Predicted to stay	Predicted to leave
Stayed	1,863 correctly identified	10 incorrectly flagged
Left	28 missed	351 correctly identified

These numbers reveal the real operational trade-off. The model identified 351 customers who could be considered for early outreach, while 28 actual churners would not have entered the model-led priority queue. Ten customers who stayed would have been flagged unnecessarily. The appropriate response is to combine targeted outreach with good service for everyone, rather than rely on predictions as the only form of customer care.

Comparison with the first milestone

Reported measure	Milestone 1: XGBoost	Milestone 2: Extra Trees
Accuracy	96%	98.31%
Recall	94%	92.61%
Precision	86%	97.23%
F1 score	90%	94.86%

The reported precision and F1 are higher in Milestone 2, while reported recall is slightly lower. This is a meaningful caution: the new model does not outperform the old one on every metric. Moreover, the milestone figures should not be treated as a controlled head-to-head test unless both models are evaluated on exactly the same customers, with matching definitions and decision thresholds.

5. From model performance to customer decisions

A churn prediction becomes valuable only when it changes an appropriate business action. ChurnGuard provides a way to assess individual accounts, score batches of customer records, group results into risk bands and export them for review. The automated workflow can process new or modified CSV/Excel inputs and send alerts for newly identified High/Critical-risk customers.

The deployed model is held at models/churnguard_best_model.joblib, with its input specification in models/feature_schema.json. These files support consistent predictions across the application and batch workflow. The separate alert histories used by the application and automation should eventually be consolidated into a shared, auditable system before broader operational use.

Recommended operating procedure

- Identify: Review newly flagged higher-risk accounts after each scoring cycle.
- Validate: Check recent complaints, customer value, contact permissions and service history before acting.
- Respond: Start with personalised service recovery; reserve costly incentives for cases with a sound business justification.
- Record: Log the contact, cost, offer, customer response and subsequent outcome.
- Learn: Review which actions work and whether the model remains reliable over time.

A prediction is not a diagnosis. A high score does not explain why a particular customer may leave, nor does it identify which offer would persuade that person to stay. Staff judgement and evidence from actual interventions remain essential.

6. Financial relevance and the retention business case

Milestone 1 calculated probability-weighted monthly revenue exposure, using customer revenue multiplied by estimated churn probability. Its reported 372 priority accounts and 78.60% exposure share illustrate how concentration can guide retention resources. However, these figures were generated by the earlier XGBoost model. They must be recomputed using current Extra Trees predictions before being described as current business exposure.

The same caution applies to Milestone 1 financial scenarios. Assumed success rates, intervention costs and projected net benefit are useful planning inputs, but they are not observed savings. No verified intervention uplift or realised return on investment has yet been established.

How management should evaluate financial value

For each customer, estimate potential revenue exposure using the current model and a defined revenue period. Next, consider the customer’s contribution margin, the probability that outreach will make a difference and the full cost of the intervention. The final decision should be based on expected incremental contribution, not on churn probability alone.

For example, an account producing 120 units of monthly revenue with a predicted churn probability of 75% has a probability-weighted monthly exposure of 90 units. This is only an illustration of prioritisation:

it does not mean 90 units will certainly be lost or that 90 units can be saved. The model's probability calibration should be checked before treating risk estimates as reliable financial weights.

Retention options

The initial report considered personalised outreach, loyalty rewards, vouchers and premium benefits, with illustrative cost assumptions of 5%, 10%, 15% and 60%. These can inform the design of an experiment, but the real costs and effectiveness of each intervention must be measured. A sensible policy is to try low-cost service recovery first and reserve expensive offers for customers whose expected incremental value justifies them.

7. Implementation plan and accountability

A controlled pilot is the most credible next step. Eligible higher-risk customers should be assigned, where appropriate, to a targeted outreach group and a normal-service comparison group. Both groups should be observed over the same period, with consistent definitions of churn and retention. The purpose is to determine whether intervention changes outcomes beyond what would have happened anyway.

Stage	Action	Owner / decision
Prepare	Confirm current model, data quality, contact permissions and scoring schedule.	Data Science and Operations
Pilot	Select eligible accounts; agree outreach rules and comparison design.	Marketing and Operations
Measure	Track actual churn, costs, customer feedback and incremental contribution.	Finance and Analytics
Decide	Scale, revise or stop based on pre-agreed commercial criteria.	Executive sponsor

Measures of success

- Observed churn: The share of customers who actually leave in each pilot group.
- Incremental retention: The difference in customers retained because of the intervention, not merely customers who were contacted.
- Cost per additional customer retained: Total intervention cost divided by the extra customers retained.
- Net incremental contribution: Additional contribution margin less outreach, incentive and administrative costs.
- Model reliability: Precision, recall and probability calibration on new customer outcomes.

Leadership should agree the minimum worthwhile retention improvement and maximum affordable cost before launching the pilot. Without these decision rules, a successful prediction model can still lead to an uneconomic retention programme.

8. Risks, limitations and governance

The strong test results are encouraging but should not be mistaken for guaranteed future performance. The evaluation reflects one historical dataset; changes in products, customer behaviour or the economy may reduce accuracy. Scores also do not establish causation, and no retention experiment has yet demonstrated incremental revenue or profit.

Operational safeguards are equally important. Customer data and contact details require controlled access; email credentials should be kept out of source code; alerts and actions should be recorded centrally; and contact preferences must be respected. The team should monitor model behaviour across relevant customer groups, investigate unexpected differences and periodically reassess both data quality and predicted risk calibration.

For reproducibility, the final notebook, model artifact, feature schema and deployment documentation should remain aligned.

9. Conclusions and recommendations

Milestone 1 established the commercial opportunity: churn affected a meaningful minority of customers, estimated exposure was concentrated, and an XGBoost-powered application demonstrated a workable early-warning concept. Milestone 2 strengthened the analytical basis by comparing nine approaches, tuning their settings, selecting Extra Trees using five-fold validation and checking its performance on a separate test sample.

The final model identified 351 of 379 churners in the test group, with only 10 customers incorrectly flagged. This makes ChurnGuard a promising decision-support capability for focused customer care. It does not replace business judgement, guarantee that a customer can be saved or prove that retention spending will generate a return.

Recommended priorities

- First: Recalculate current risk groups and revenue exposure using the selected Extra Trees model.
- Second: Establish a single process for assigning, contacting and recording higher-risk accounts.
- Third: Conduct a controlled pilot with low-cost initial interventions and an appropriate comparison group.
- Fourth: Evaluate incremental retention and contribution after all intervention costs.
- Finally: Monitor model reliability and update the approach when new evidence warrants it.

Final business assessment: ChurnGuard has progressed from an exploratory prediction concept to a stronger, more repeatable customer-risk system. Its potential value lies in enabling earlier and better-prioritised customer contact. The next milestone for the business is to demonstrate that this action actually reduces churn and produces sustainable financial benefit.

Evidence basis

Milestone 1: ChurnGuard — Customer Churn Prediction & Retention Strategy, 15-page report supplied for this review. Milestone 2: modelling results and deployment details reported in the project notebook and preceding Milestone 2 report. Figures from the two milestones are kept distinct; scenarios are labelled illustrative where no observed business outcome is available.